

1.3 Competitive Event: The Heuristic Red-Team Duel

- **Phase A: Blue Team Defense (15 min):** All teams design a manual rule-based decision program using at most **two** conditional branches on individual pixel intensities (e.g., `if (b_00 <= theta_1 and b_11 >= theta_2) return 1 else 0`) that achieves 100% accuracy on $\mathcal{D}$. Submit your rule to the instructor.
- **Scoring:** +100 AP if your rule correctly separates $\mathcal{D}$.

- **Phase B: Red Team Adversarial Strike (15 min):** Squads swap rules (confirm your opponent squad with instructor). Your squad receives the opposing team's heuristic. Craft a single synthetic wafer $\mathbf{x}_{\text{adv}} \in [0, 255]^4$ with an obvious physical defect (e.g., severe light blockage where nominal wafers require light, or high diagonal glare) that **breaks** their rule (causes a false positive or false negative).
- **Scoring:** +150 AP if your adversarial sample breaks the opponent's heuristic while remaining physically plausible.